Introduction to R Workshop

Getting Started, Data Visualization, and Regression

Muji Chughtai

2026-07-20

Getting Started

Introduction to R and RStudio

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- RStudio also allows for powerful document creation using Markdown, which can generate reports, presentations, and dashboards (this presentation was made using a Quarto Markdown in RStudio).

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4. **Files/Plots/Packages/Help/Viewer Panel (Bottom Right)**: This panel has multiple tabs for managing files, viewing plots, installing and loading packages, accessing R help documentation, and displaying web content.

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  - If you want to write in other languages in the code chunk, you can specify the language in the curly braces (e.g., `{python}`, `{bash}`, etc.).
- I suggest you do your analyses in R Markdown documents, as it allows you to keep your code and results together in a reproducible format that can be easily exported to various formats (HTML, PDF, Word, etc.)!

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- The `here` package is a great tool to help you manage file paths in your R projects. It allows you to use relative paths instead of absolute paths, which makes your code more portable and easier to share.

Data Visualization

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- Open RStudio and create a new project (and corresponding folder) by navigating from the menu to `File > New Project > New Directory > New Project`, and name your project whatever you like and save it in a location you prefer.
- Within that project folder, create a new R Markdown document by navigating to `File > New File > R Markdown`, and you can do this from either the menu or the bottom right panel. Name it whatever you like!

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library(tidyverse)
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- You can also directly install packages from the console by typing in the same code above (when you run a chunk it will run in the console).

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- These headings will transfer into the output document that you render for easy navigation in the html, pdf, or word document.

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- The data and codebook is available on my github at https://github.com/mujtabachughtai/2026-R-Workshop.
- Download the data (called `WB.2016.csv`), create a `Data` subfolder in this project's directory, and save the data in that folder.

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- Note: You normally don't need the `fileEncoding` argument, but this dataset has some special characters that require it.
- You should now be able to view your data in the `Environment` tab in the top right panel of RStudio.

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  - `Population`: The population of the country

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  - `Population`: The population of the country
  - `GNI`: Gross National Income per capita
- And we're going to make some variables of our own, too.

Getting a Feel for the Data

- Let's take a look at the first few rows of these key variables. We will use the `knitr::kable` and `head` functions to do so.

```
knitr::kable(head(
  WorldBank[, c("Country", "Code", "Population", "GNI")]),
  caption = "First 5 Rows of our Key Variables from the
2016 World Bank Data")
```

Getting a Feel for the Data

Table 1: First 5 Rows of our Key Variables from the 2016 World Bank Data

Country	Code	Population	GNI
Afghanistan	AFG	34656032	580
Albania	ALB	2876101	4320
Algeria	DZA	40606052	4360
American Samoa	ASM	55599	NA
Andorra	AND	77281	NA
Angola	AGO	28813463	3450

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mean(WorldBank$Population, na.rm = TRUE)
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# mean of GNI
mean(WorldBank$GNI, na.rm = TRUE)
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```
ggplot(aes(x = Population), data = WorldBank) +
  geom_density()
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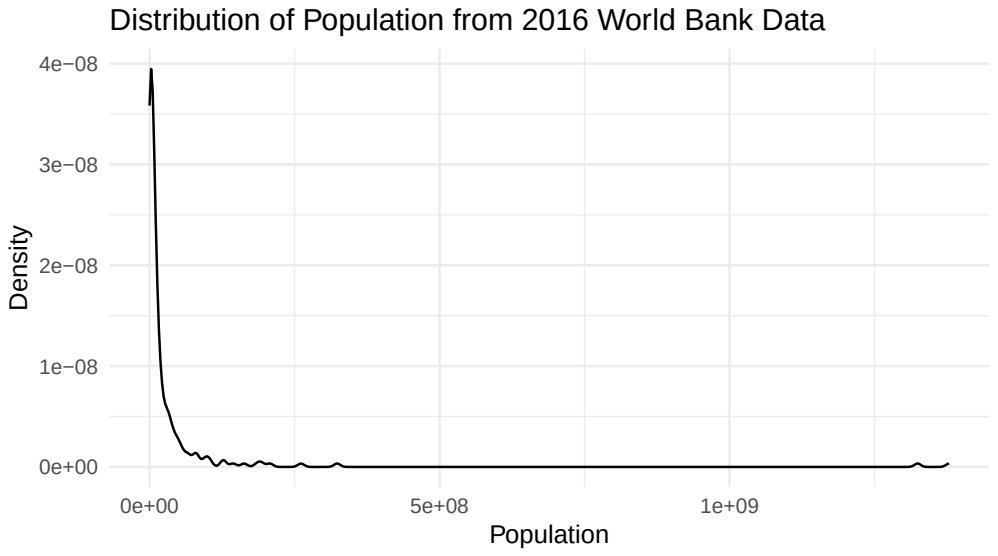
```
WorldBank %>%
  ggplot(aes(x = Population)) +
  geom_density() +
  theme_minimal() +
  labs(
    title = "Distribution of Population from 2016 World Bank Data",
    x = "Population",
    y = "Density"
  )
```

Getting a Feel for the Data



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  - This is because the vast majority of countries that have small populations, while there are a few that have very large ones.
- We can log-transform the data to make it more normally distributed, which is a common technique in statistics. This will also help us with our regression analyses later on.

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- The `mutate` function allows you to create new variables based on the data from other variables. Let's use this function to create a new variable called `logPopulation` that is just the natural log of the `Population` variables, and we will recreate the `WorldBank` dataframe with this new variable:

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WorldBank <- WorldBank %>%
  mutate(logPopulation = log(Population))
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WorldBank %>%
  ggplot(aes(x = logPopulation)) +
  geom_density() +
  theme_minimal() +
  labs(
    title = "Distribution of
log-transfored Population from 2016 World Bank Data",
    x = "log(Population)",
    y = "Density"
  )
```

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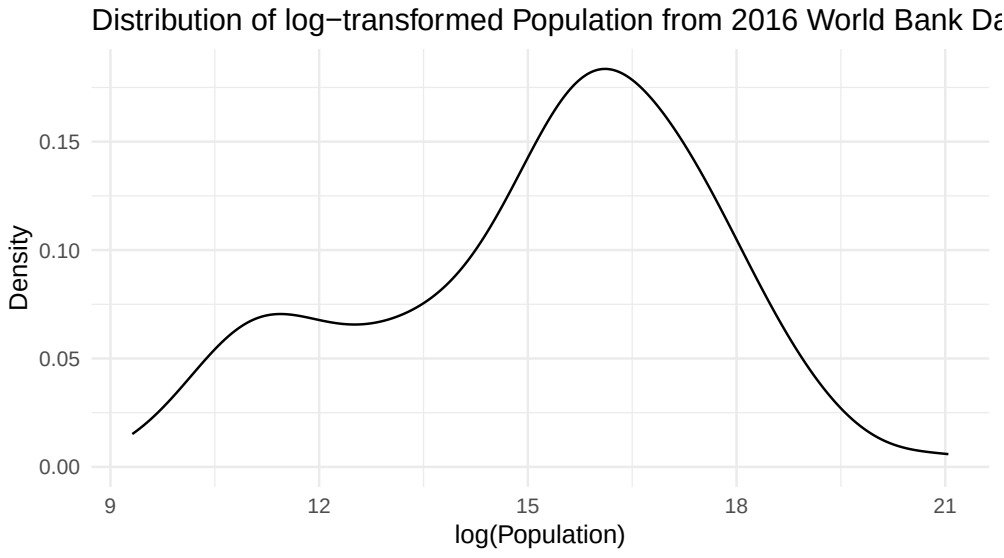

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```
WorldBank <- WorldBank %>%
  mutate(logGNI= log(GNI))

WorldBank %>%
  ggplot(aes(x = logGNI)) +
  geom_density() +
  theme_minimal(base_size = 22) +
  labs(
    title = "Distribution of log-transformed
GNI from 2016 World Bank Data",
    x = "log(GNI)",
    y = "Density"
  )
```

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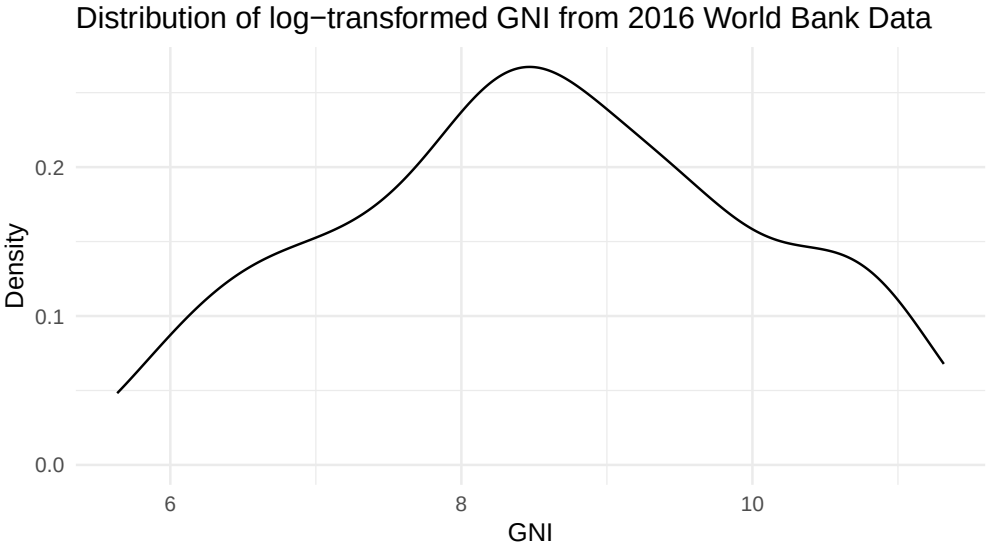

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WorldBank %>%
  ggplot(aes(x = logGNI, y = logPopulation)) +
  geom_point(alpha = 0.5) +
  theme_minimal(base_size = 22) +
  labs(
    title = "Country GNI vs Population from 2016 World Bank Data",
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    y = "log(GNI)"
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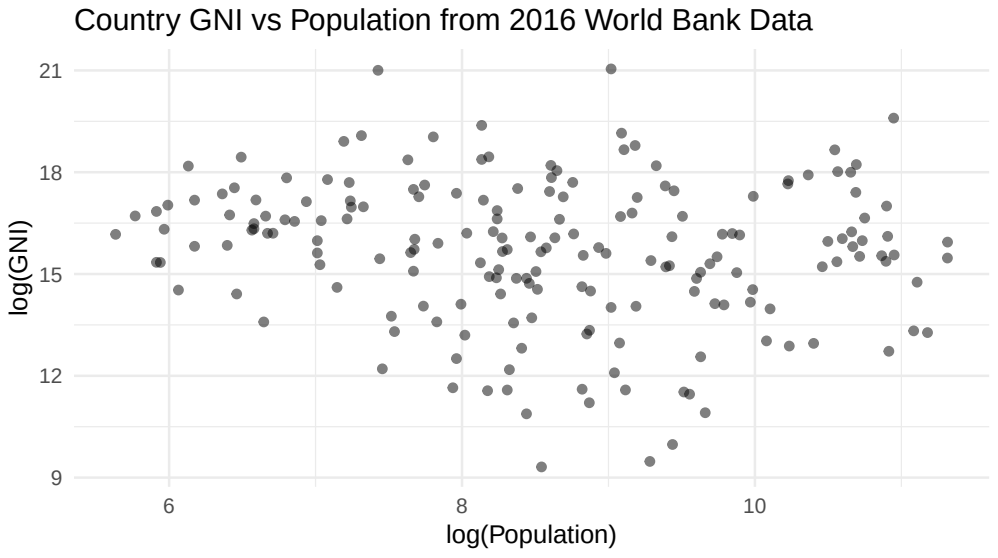

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  geom_point(alpha = 0.5) +
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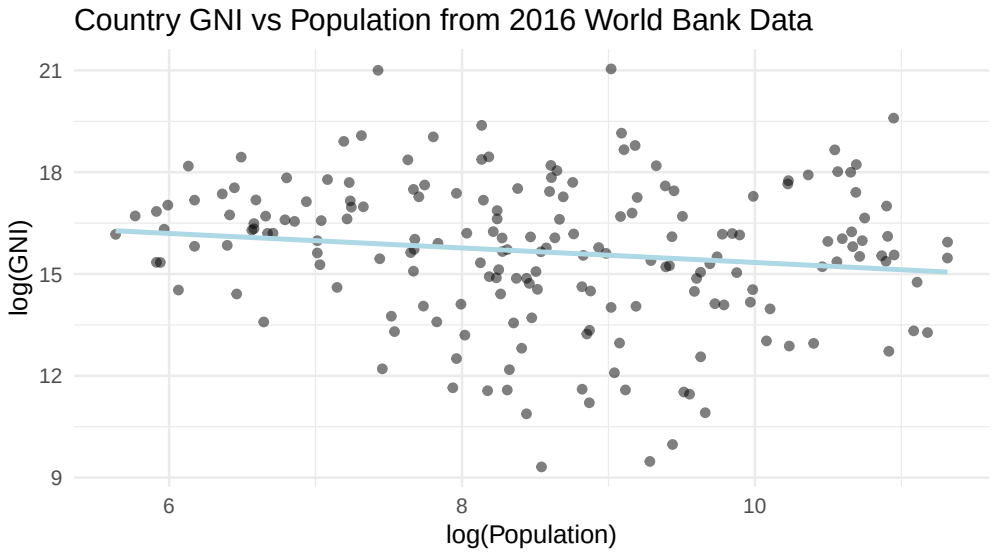
```

Plotting Associations



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- Specifically, we can color countries based on the continent that they come from to get an initial sense on if the relationship between a country's population and it GNI might differ across different continents.
- However, when looking at the current `WorldBank` dataframe, there is no continent variable.
- We could use `mutate` again to create this new `continent` variable, but there are too many countries to assess the value of to create the corresponding continent.

Combining Dataframes

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- To combine dataframes, you have to have a variable that is shared in both so that rows are combined when the values in that column are shared.
- For us, that would be the `Country` variable
- Let's load in the `CountryContinents.csv` file into our workspace and combine it into a new `WorldBank` dataframe using `left_join`

Combining Dataframes

```
CountryContinents <- read.csv(
  here("Data", "CountryContinents.csv"),
  fileEncoding = "Windows-1252"
)

WorldBank <- WorldBank %>%
  left_join(CountryContinents, by = "Country")
```

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- With this new grouping variable, we can also calculate summary statistics for each group separately. Let's do this before remaking the `logPopulation` vs `logGNI` plot with `Continent`.
- To calculate summary statistics for each group, we can use the `group_by` and `summarise`:

Computing Summary Statistics

```
WorldBank %>%
  group_by(Continent) %>%
  summarise(
    n = n(),
    mean_logGNI = mean(logGNI, na.rm = TRUE),
    mean_logPopulation = mean(logPopulation, na.rm = TRUE)
  ) %>%
  knitr::kable(caption = "Summary Statistics by Continent",
               col.names = c("Continent", "Number of Countries",
                             "Mean log(GNI)", "Mean log(Population)"))
```

Computing Summary Statistics

Table 2: Summary Statistics by Continent

Continent	Number of Countries	Mean log(GNI)	Mean log(Population)
Africa	54	7.198026	16.03096
Asia	67	8.685735	15.32228
Europe	49	9.803978	14.93849
North/Central America	34	9.045212	13.52896
South America	12	8.842763	16.33919

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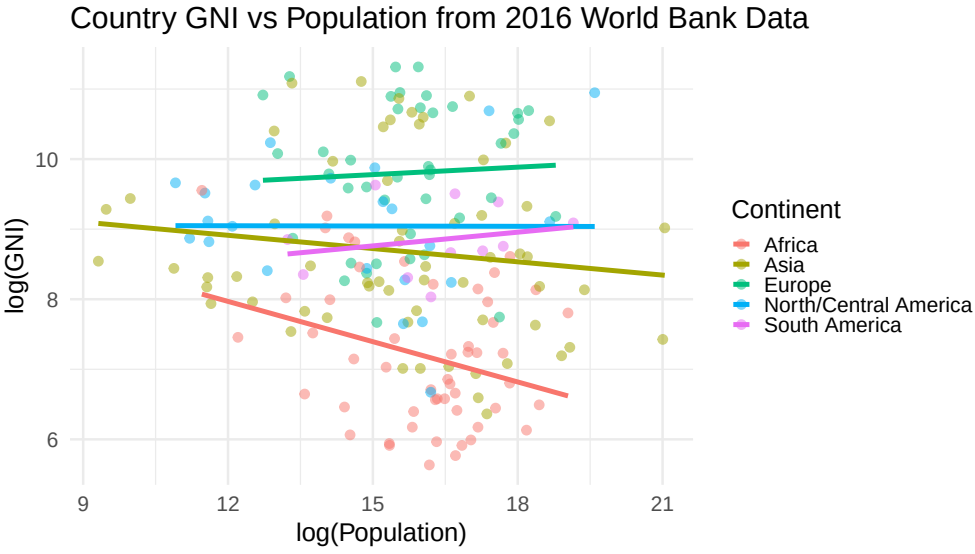
```
WorldBank %>%
  filter(!is.na(Continent)) %>%
  ggplot(aes(x = logPopulation, y = logGNI, color = Continent)) +
  geom_point(alpha = 0.5) +
  theme_minimal(base_size = 22) +
  labs(
    title = "Country GNI vs Population from 2016 World Bank Data",
    x = "log(Population)",
    y = "log(GNI)"
  )
```

Plotting 3 Variables



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```
WorldBank %>%
  filter(!is.na(Continent)) %>%
  ggplot(aes(x = logPopulation, y = logGNI)) +
  facet_wrap(~Continent) +
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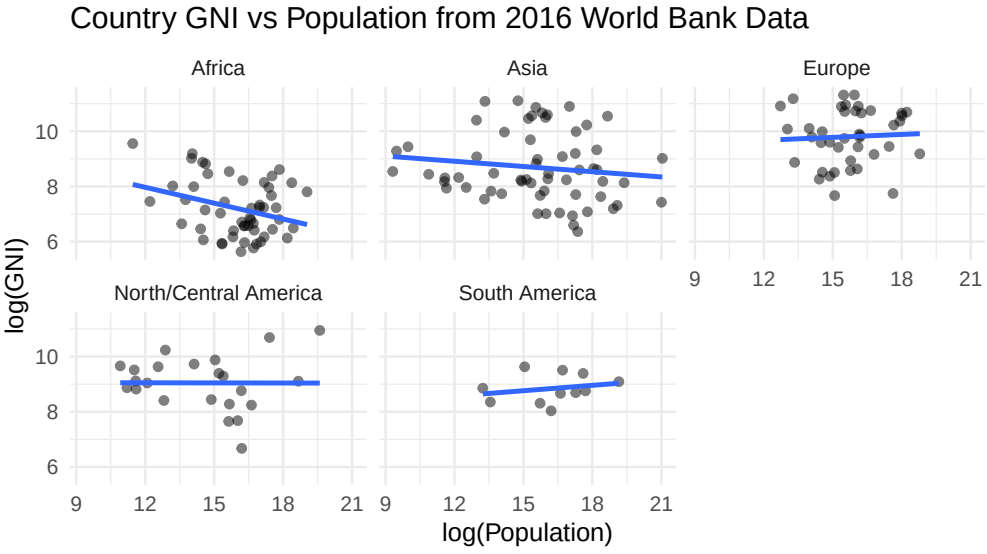

Saving Plots

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- This is a nice plot! We can save it to our `Figures` folder in our directory (make sure you have a folder named `Figures` in your working directory) by saving the `ggplot` object and using `ggsave`:

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```
GNI_Population_by_Continent <- WorldBank %>%
  filter(!is.na(Continent)) %>%
  ggplot(aes(x = logPopulation, y = logGNI)) +
  facet_wrap(~Continent) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  theme_minimal(base_size = 22) +
  labs(
    title = "Country GNI vs Population from 2016 World Bank Data",
    x = "log(Population)",
    y = "log(GNI)"
  )
```

Saving Plots

```
ggsave(
  GNI_Population_by_Continent,
  filename = here("Figures", "GNI_Population_by_Continent.png"),
  width = 32,
  height = 24,
  units = "in",
  dpi = 300
)
```

Linear Regression

Theory

Definition

Ordinary least squares regression estimates the intercept and slope by minimizing the sum of the squared residuals:

$$(\hat{\beta}_0, \hat{\beta}_1) = \underset{\beta_0, \beta_1}{\arg\min} \sum_{i=1}^{n} \left[y_i - (\beta_0 + \beta_1 x_i) \right]^2 .$$

The resulting regression equation is

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i .$$

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- Given that it seems like the relationship between a country's GNI and its population might differ across continents, we will fit an interaction model.
  - This will also take into account the likely main effect of `Continent` on our other two variables.

Implementation

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```
FullModel <- lm(logGNI ~ logPopulation * Continent, data = WorldBank)
summary(FullModel)
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- And we can further examine interactions by using the `simple_slopes` function:

```
library(reghelper)
simple_slopes(FullModel, pred = "logPopulation", modx = "Continent")
```

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  - Even after controlling for the main effect of `Continent`, there is still a significant negative relationship between a country's population and its GNI.
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  - No interactions are significant: the slope does not significantly change from this reference group for any of the other continents.

Interpretation

- From this output, we can make a few interpretations
  - Even after controlling for the main effect of `Continent`, there is still a significant negative relationship between a country's population and its GNI.
  - When Africa is the reference group, the slope of the relationship between a country's population and its GNI is -0.19
  - No interactions are significant: the slope does not significantly change from this reference group for any of the other continents.
  - However, when examining the slopes for different continents, the association between a country's population and its GNI is only significant for Africa.

One Last Fun Plot

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- You can add a lot of different variables to `ggplot` object, but at risk of making the plots hard to interpret (there are only so many ways to manipulate a visual object!).

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- You can add a lot of different variables to `ggplot` object, but at risk of making the plots hard to interpret (there are only so many ways to manipulate a visual object!).
- Let's try to add a bit for information to our plot by changing the points to the 3-character country code and by changing the color of the points based on how much fossil fuels are used in the country (as a percentage of total energy use).

Plotting 4 Variables

```
WorldBank %>%
  filter(!is.na(Continent)) %>%
  ggplot(aes(x = logPopulation, y = logGNI, color = FossilPct)) +
  geom_text(aes(label = Code), size = 2) +
  facet_wrap(~Continent) +
  geom_smooth(method = "lm", se = FALSE) +
  theme_minimal(base_size = 14) +
  labs(
    title = "Country GNI vs Population from 2016 World Bank Data",
    x = "log(Population)",
    y = "log(GNI)",
    color = "% of Energy from Fossil Fuels"
  )
```

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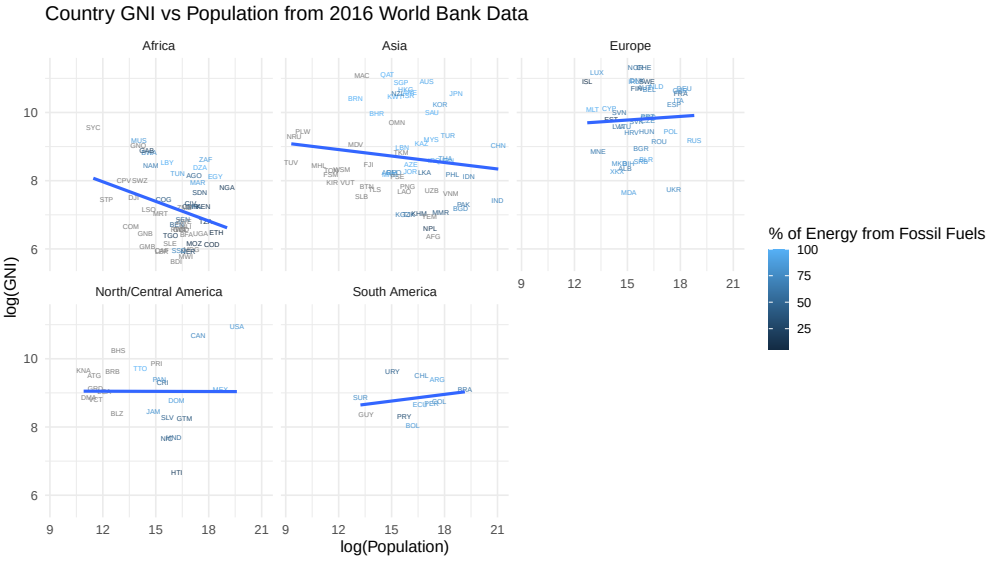

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Pop_GNI_Fossil <- WorldBank %>%
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  geom_text(aes(label = Code), size = 2) +
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```

Plotting 4 Variables

```
ggsave(
  Pop_GNI_Fossil,
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  units = "in",
  dpi = 300
)
```

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